E-commerce sales dataset

Assignment Description

This assignment focuses on data cleaning, data transformation, data merging, aggregation, and data modelling using Power BI. It helps to prepare accurate data and analyse sales, profit, and sales targets.

Column Description

List of Orders

Column Name	Simple Description
Order ID	Order number.
Order Date	Date of the order.
Customer Name	Customer's name.
City	Customer's city.
State	Customer's state.
Category	Product category.
Amount	Total sales amount.
Profit	Profit or loss amount.
Location	City and state combined.
Profit Margin	Profit percentage.
Profit Status	Shows Profit, Loss, or Break-Even.

Order Details

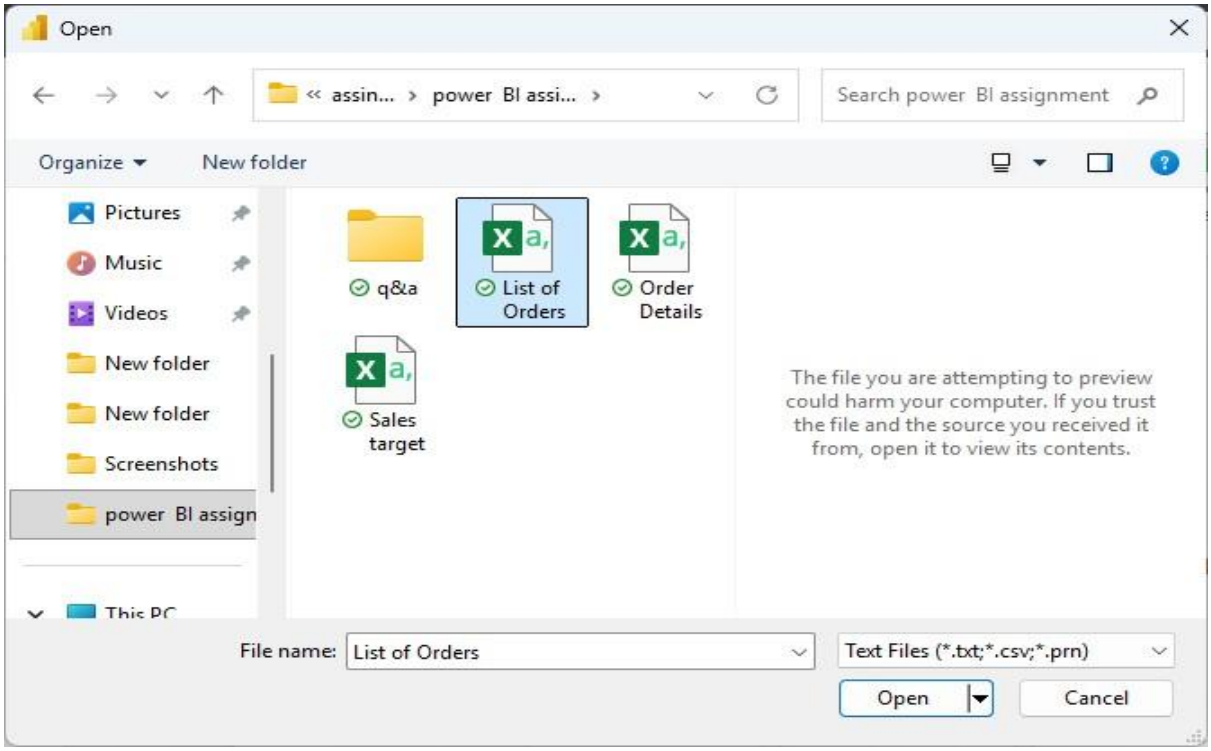
Column Name	Simple Description
Order ID	Order number.
Category	Product category.
Sub-Category	Product type.
Quantity	Number of products sold.
Amount	Sales amount.
Profit	Profit or loss amount.

Sales Target

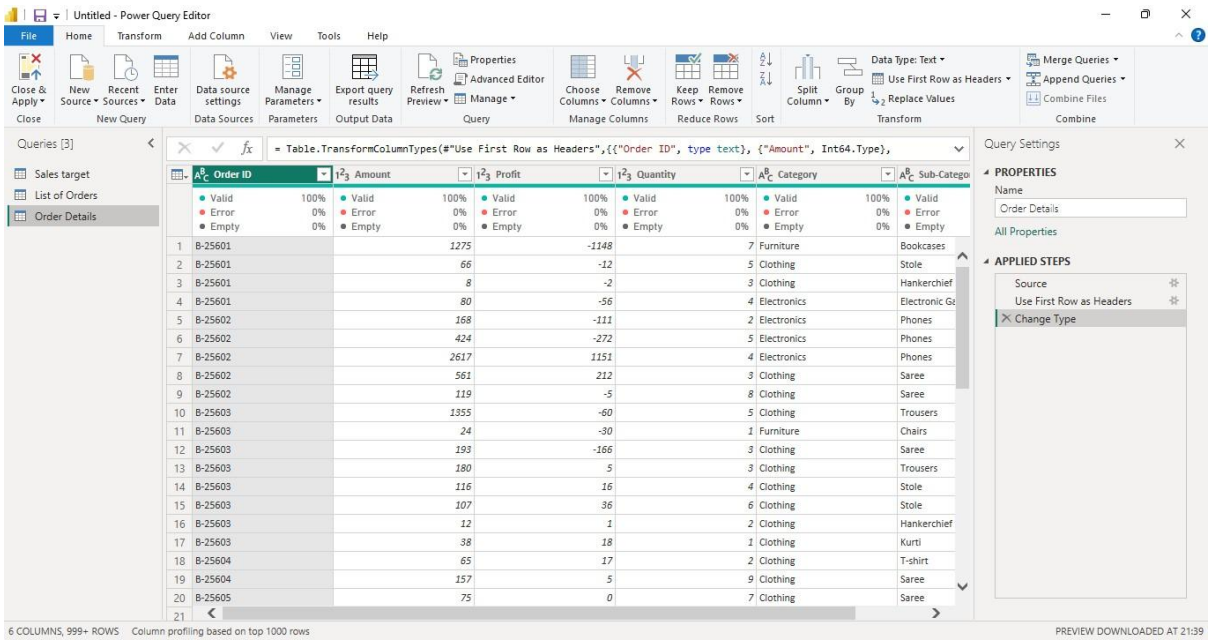
Column Name	Simple Description
Category	Product category.
Month of Order Date	Order month.
Target	Expected sales amount.

Tasks Data Import & Transformation

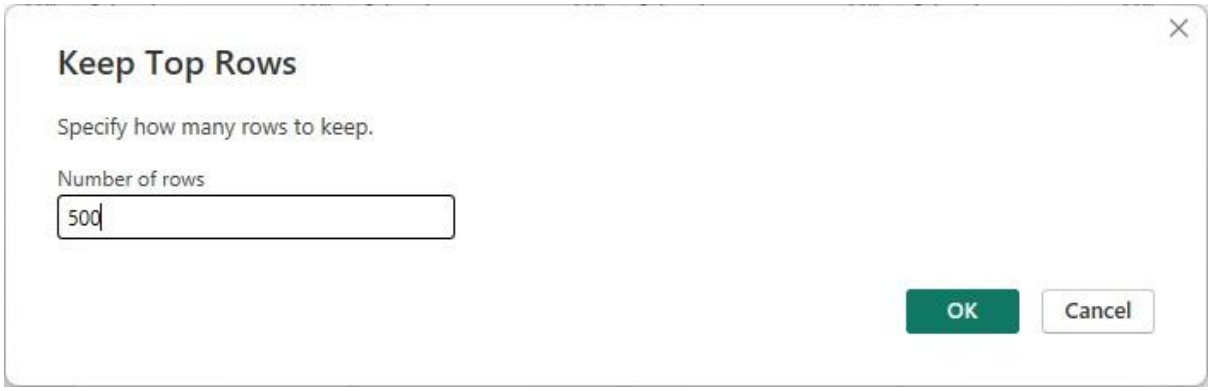
1.A) Import List of Orders.csv , Order Details.csv and Sales Target.csv into Power Query Editor:



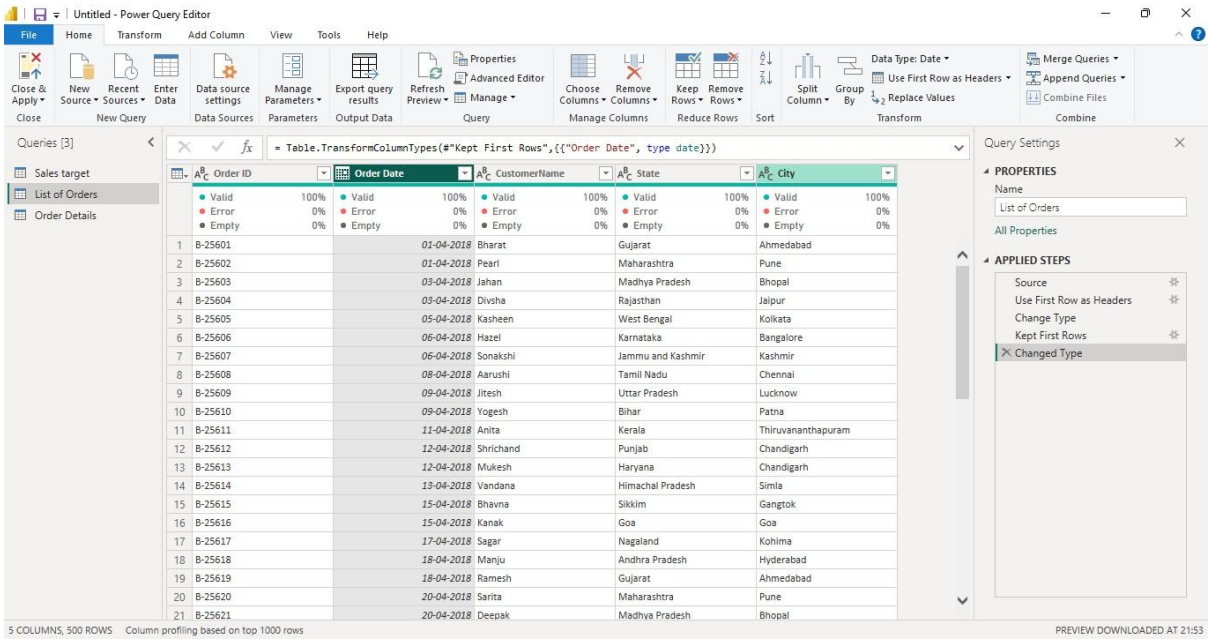
1.B) Open the list of orders in the Power Query Editor using the Transform Data option:



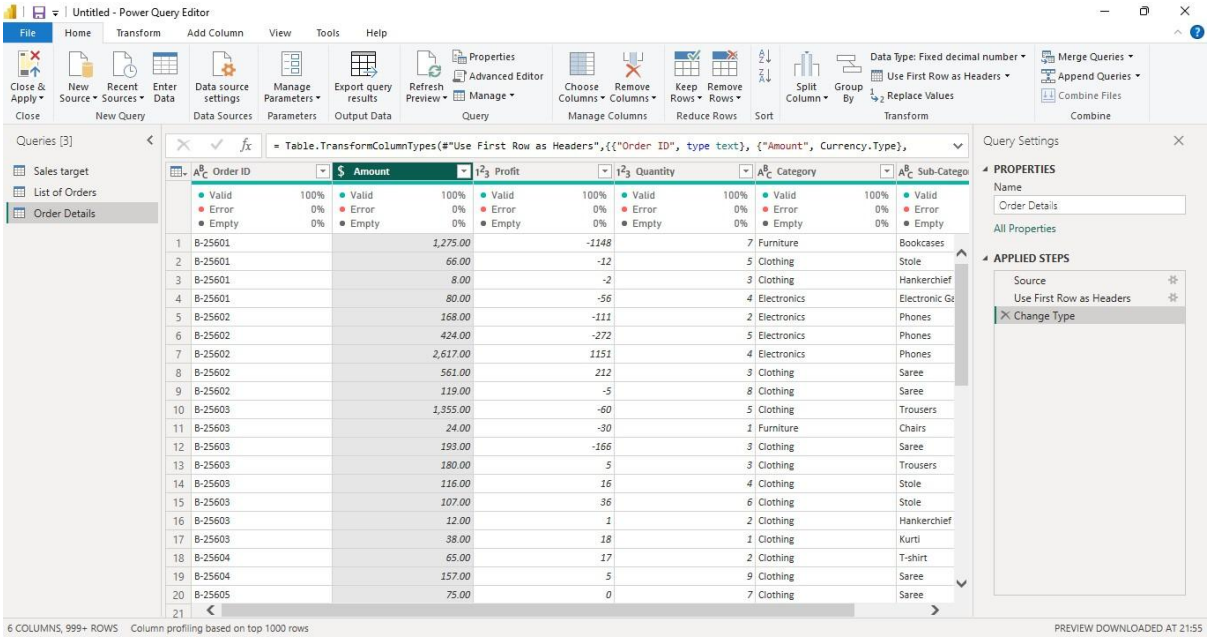
2.A) Restrict the List of Orders table to the first 500 rows:

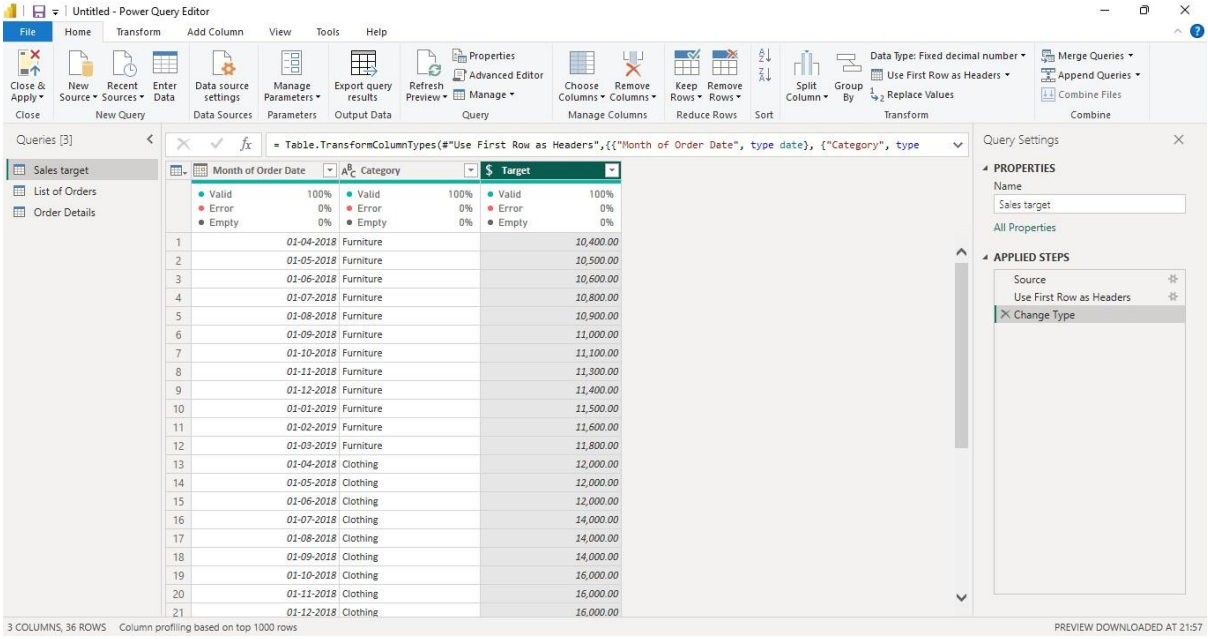


2.B) Set the Order Date column to the Date data type:

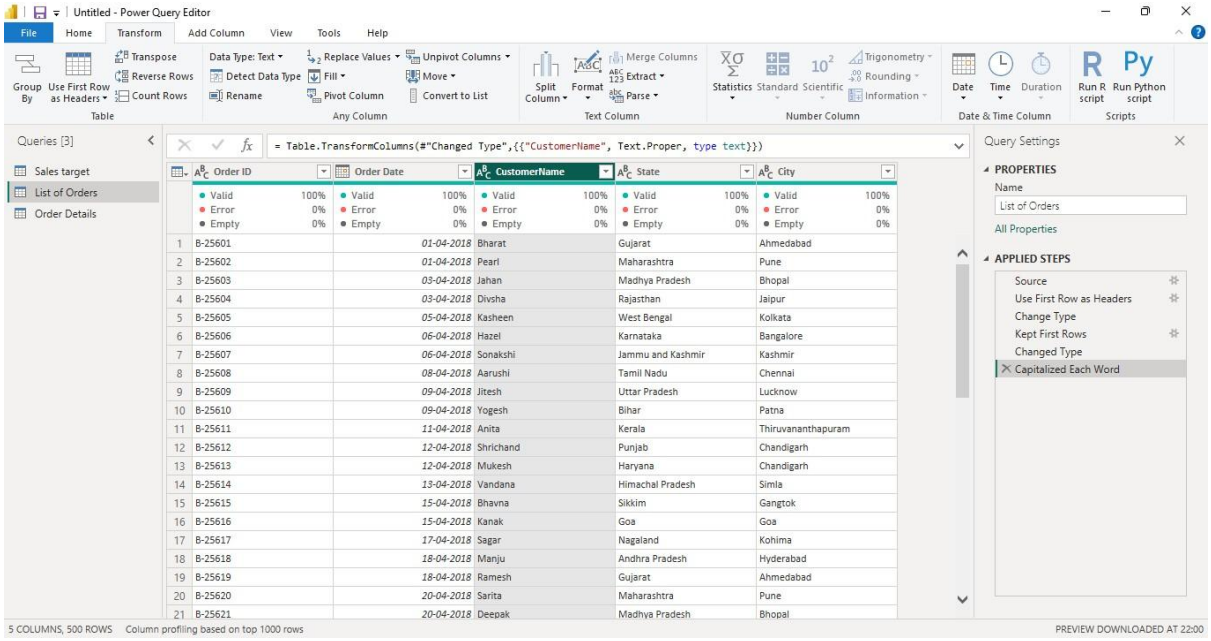


2.C) Change the Amount and Target columns to fixed decimal numbers:


Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Close & Apply
New Source
Recent Sources
Enter Data
Data source settings
Manage Parameters
Export query results
Refresh Preview
Properties
Advanced Editor
Manage
Choose Columns
Remove Columns
Keep Rows
Remove Rows
Split Column
Group By
Data Type: Fixed decimal number
Use First Row as Headers
Replace Values
Merge Queries
Append Queries
Combine Files
Close
New Query
Data Sources
Parameters
Output Data
Query
Manage Columns
Reduce Rows
Sort
Transform
Combine
Queries [3]
Sales target
List of Orders
Order Details
= Table.TransformColumnTypes(#"Use First Row as Headers",{{"Order ID", type text}, {"Amount", Currency.Type},
Order ID
Amount
Profit
Quantity
Category
Sub-Categor
Valid 100%
Error 0%
Empty 0%
1 B-25601 1,275.00 -1148 7 Furniture Bookcases
2 B-25601 66.00 -12 5 Clothing Stole
3 B-25601 8.00 -2 3 Clothing Hankerchief
4 B-25601 80.00 -56 4 Electronics Electronic Ga
5 B-25602 168.00 -111 2 Electronics Phones
6 B-25602 424.00 -272 5 Electronics Phones
7 B-25602 2,617.00 1151 4 Electronics Phones
8 B-25602 561.00 212 3 Clothing Saree
9 B-25602 119.00 -5 8 Clothing Saree
10 B-25603 1,355.00 -60 5 Clothing Trousers
11 B-25603 24.00 -30 1 Furniture Chairs
12 B-25603 193.00 -166 3 Clothing Saree
13 B-25603 180.00 5 3 Clothing Trousers
14 B-25603 116.00 16 4 Clothing Stole
15 B-25603 107.00 36 6 Clothing Stole
16 B-25603 12.00 1 2 Clothing Hankerchief
17 B-25603 38.00 18 1 Clothing Kurti
18 B-25604 65.00 17 2 Clothing T-shirt
19 B-25604 157.00 5 9 Clothing Saree
20 B-25605 75.00 0 7 Clothing Saree
21
Query Settings
PROPERTIES
Name
Order Details
All Properties
APPLIED STEPS
Source
Use First Row as Headers
Change Type
6 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 21:55

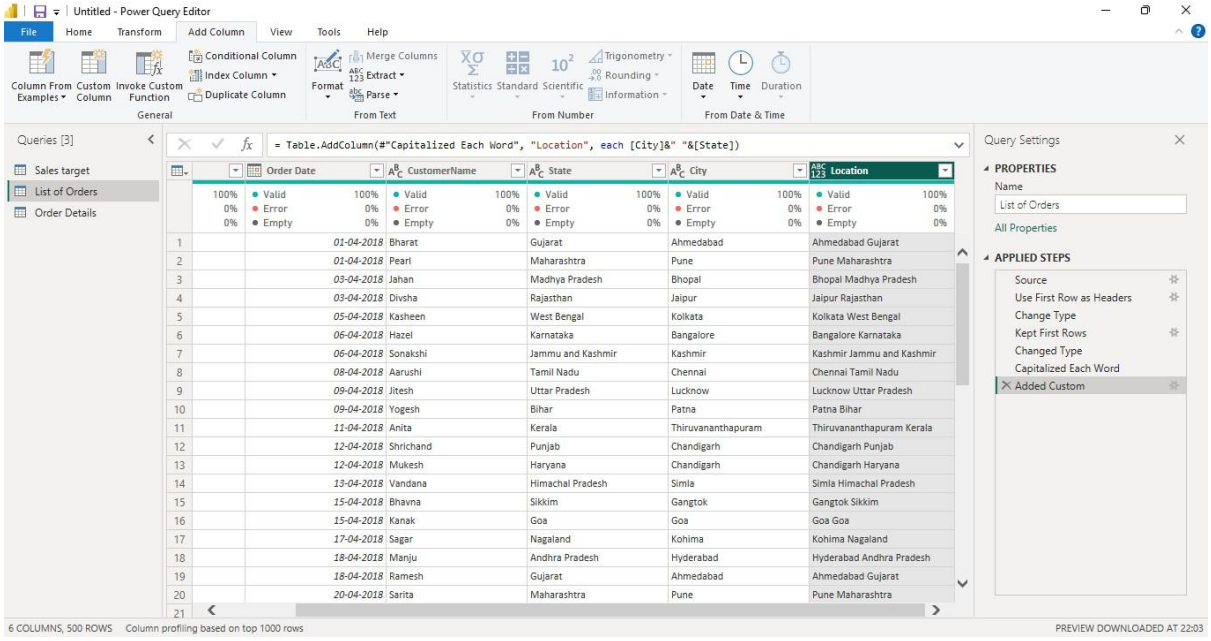

Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Close & Apply
New Source
Recent Sources
Enter Data
Data source settings
Manage Parameters
Export query results
Refresh Preview
Properties
Advanced Editor
Manage
Choose Columns
Remove Columns
Keep Rows
Remove Rows
Split Column
Group By
Data Type: Fixed decimal number
Use First Row as Headers
Replace Values
Merge Queries
Append Queries
Combine Files
Close
New Query
Data Sources
Parameters
Output Data
Query
Manage Columns
Reduce Rows
Sort
Transform
Combine
Queries [3]
Sales target
List of Orders
Order Details
= Table.TransformColumnTypes(#"Use First Row as Headers",{{"Month of Order Date", type date}, {"Category", type
Month of Order Date
Category
Target
Valid 100%
Error 0%
Empty 0%
1 01-04-2018 Furniture 10,400.00
2 01-05-2018 Furniture 10,500.00
3 01-06-2018 Furniture 10,600.00
4 01-07-2018 Furniture 10,800.00
5 01-08-2018 Furniture 10,900.00
6 01-09-2018 Furniture 11,000.00
7 01-10-2018 Furniture 11,100.00
8 01-11-2018 Furniture 11,300.00
9 01-12-2018 Furniture 11,400.00
10 01-01-2019 Furniture 11,500.00
11 01-02-2019 Furniture 11,600.00
12 01-03-2019 Furniture 11,800.00
13 01-04-2018 Clothing 12,000.00
14 01-05-2018 Clothing 12,000.00
15 01-06-2018 Clothing 12,000.00
16 01-07-2018 Clothing 14,000.00
17 01-08-2018 Clothing 14,000.00
18 01-09-2018 Clothing 14,000.00
19 01-10-2018 Clothing 16,000.00
20 01-11-2018 Clothing 16,000.00
21 01-12-2018 Clothing 16,000.00
Query Settings
PROPERTIES
Name
Sales target
All Properties
APPLIED STEPS
Source
Use First Row as Headers
Change Type
3 COLUMNS, 36 ROWS
Column profiling based on top 1000 rows
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3. Format the Customer Name column to proper case to ensure consistent capitalization :



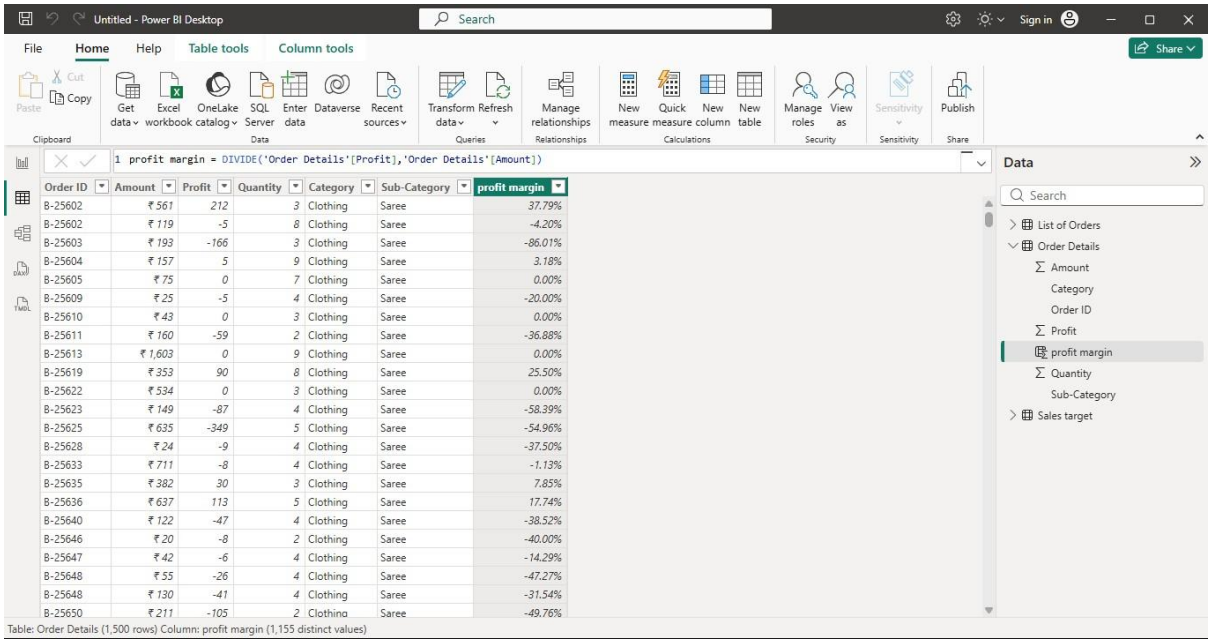
	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal

4.A) Merge the City and State columns to create a new column named Location in the format: City, State:

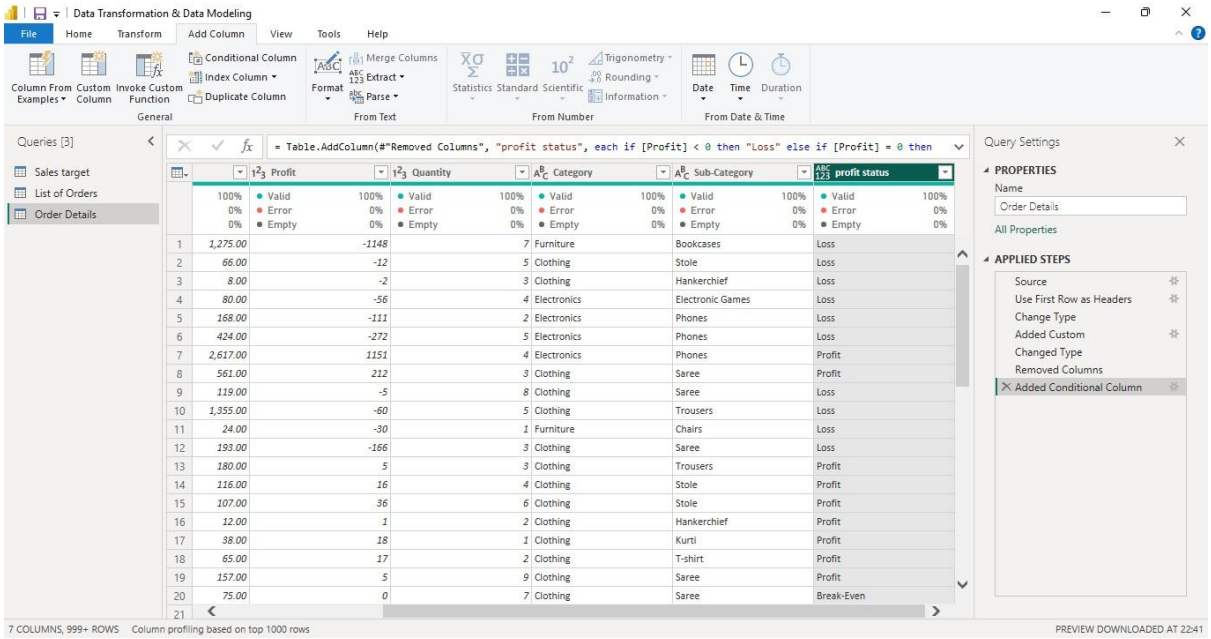


	Order Date	CustomerName	State	City	Location
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad Gujarat
2	01-04-2018	Pearl	Maharashtra	Pune	Pune Maharashtra
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal Madhya Pradesh
4	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur Rajasthan
5	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata West Bengal
6	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore Karnataka
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir Jammu and Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai Tamil Nadu
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow Uttar Pradesh
10	09-04-2018	Yogesh	Bihar	Patna	Patna Bihar
11	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram Kerala
12	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh Punjab
13	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh Haryana
14	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla Himachal Pradesh
15	15-04-2018	Bhavna	Sikkim	Gangtok	Gangtok Sikkim
16	15-04-2018	Kanak	Goa	Goa	Goa Goa
17	17-04-2018	Sagar	Nagaland	Kohima	Kohima Nagaland
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad Andhra Pradesh
19	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad Gujarat
20	20-04-2018	Sarita	Maharashtra	Pune	Pune Maharashtra

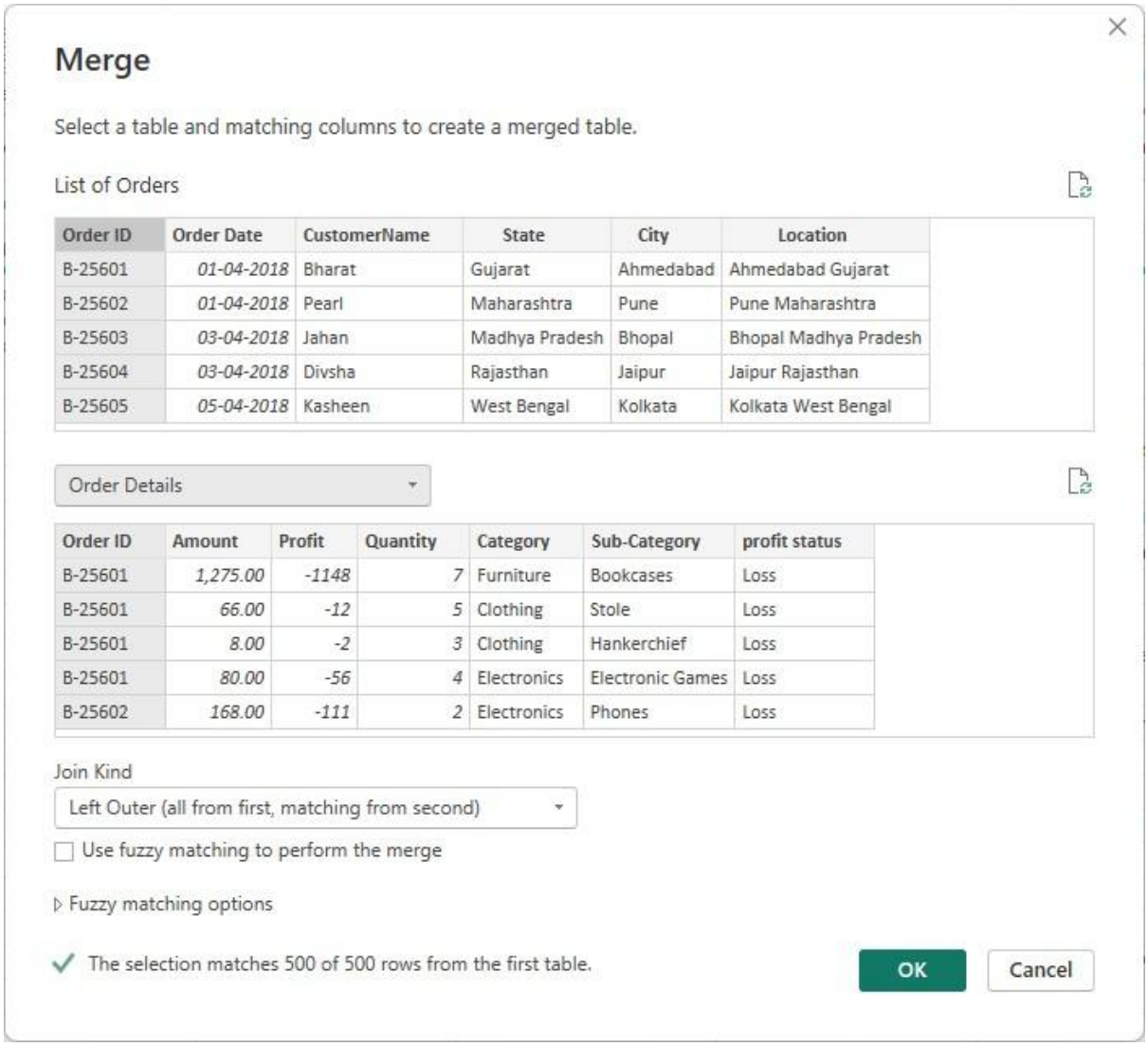
4.B) Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage):



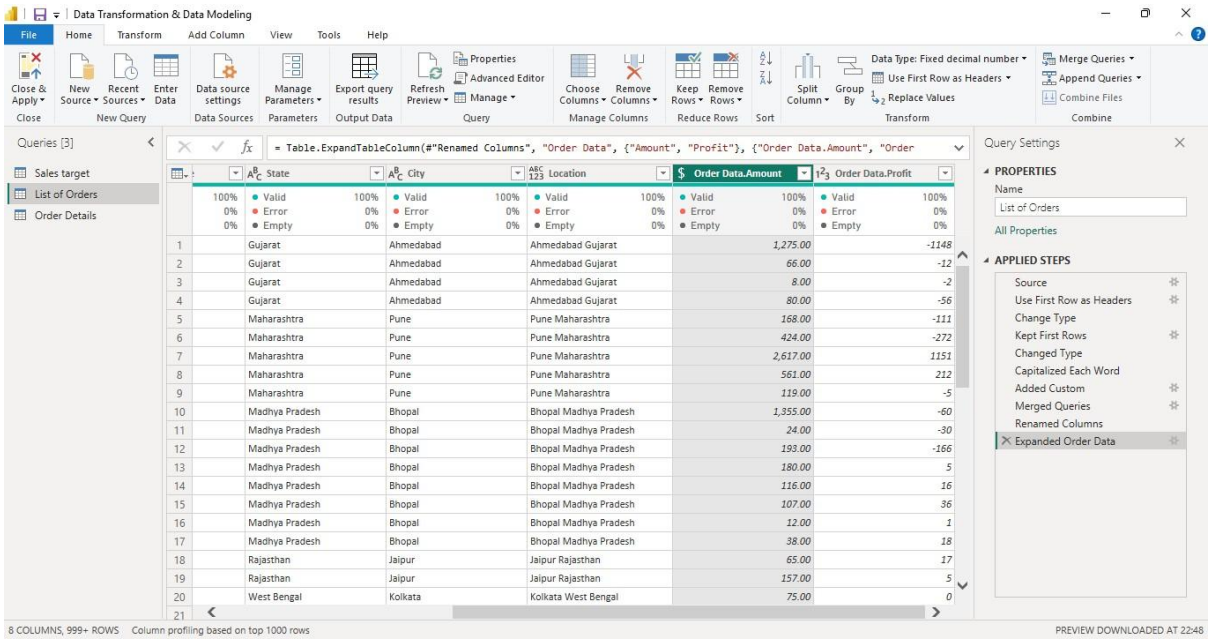
5. Add a conditional column named Profit Status based on (Profit , Loss, Break-Even) :



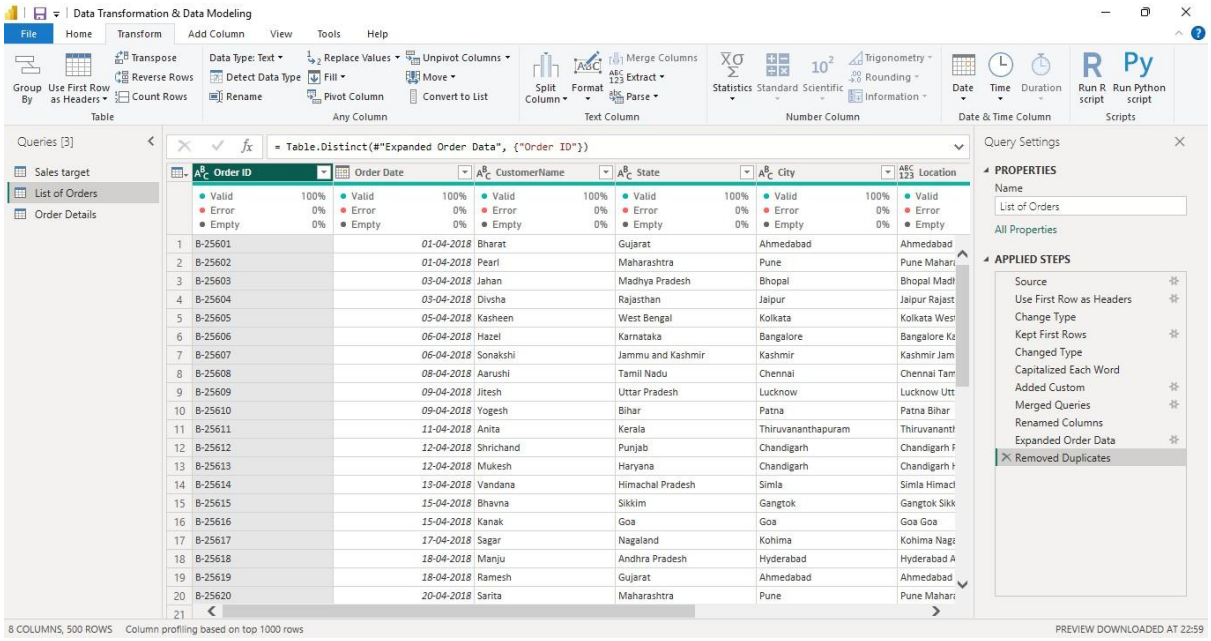
6.A) Merge the List of Orders and Order Details tables using Order ID:



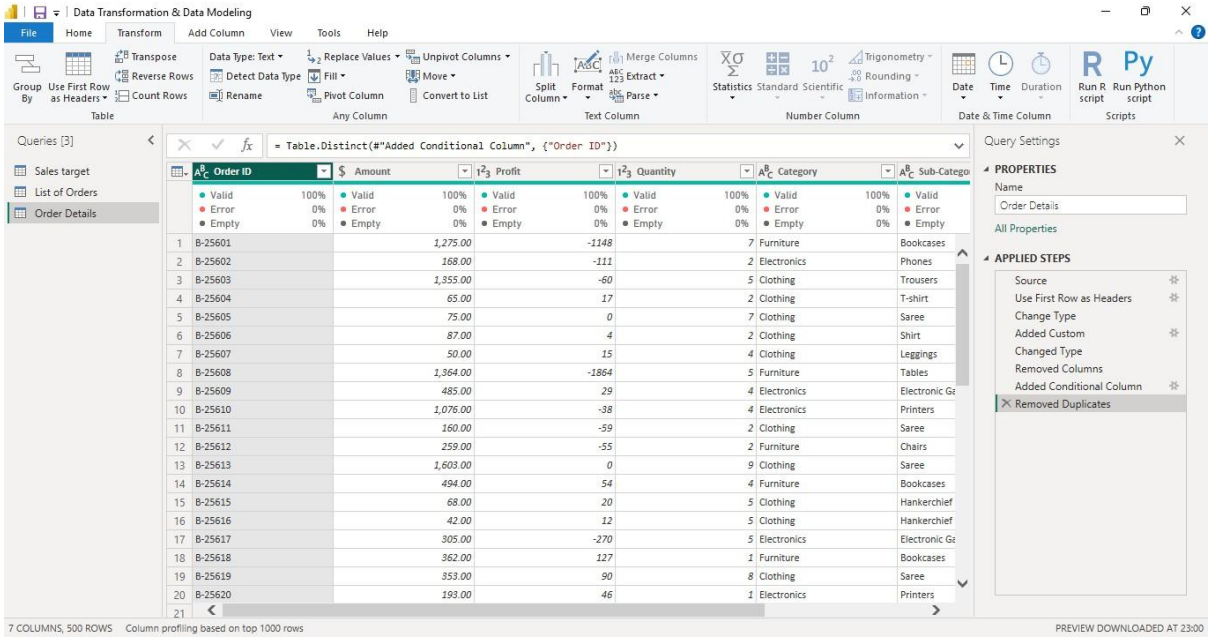
6.B) Name the merged output table as Orders Data:



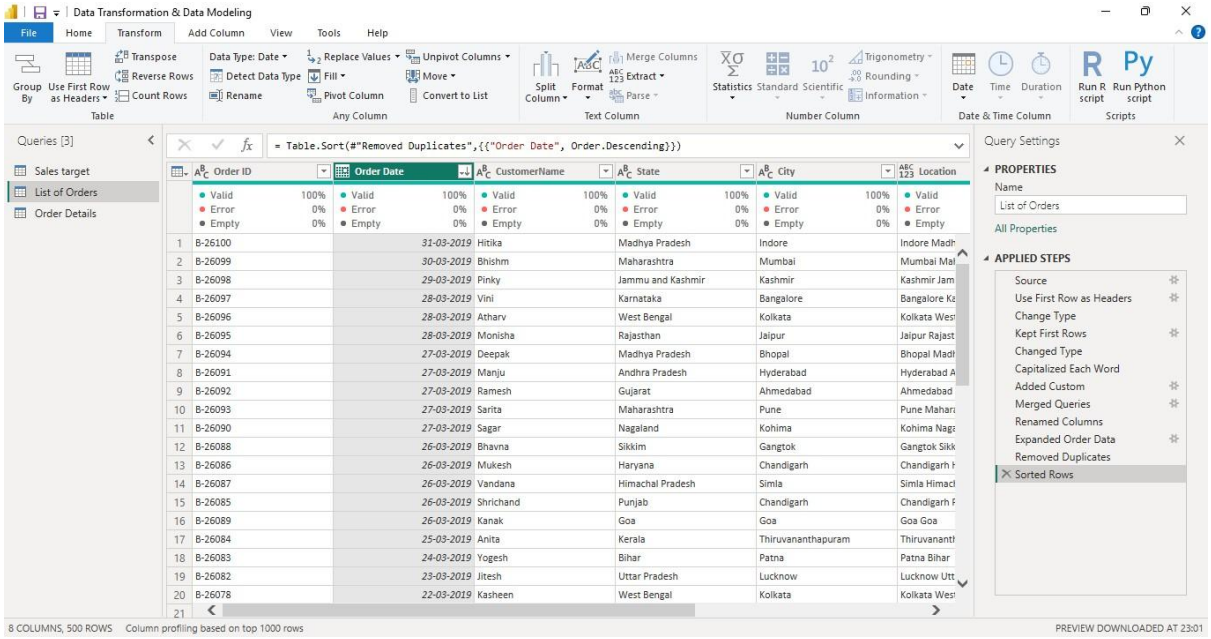
7.A) Identify missing values across all tables:



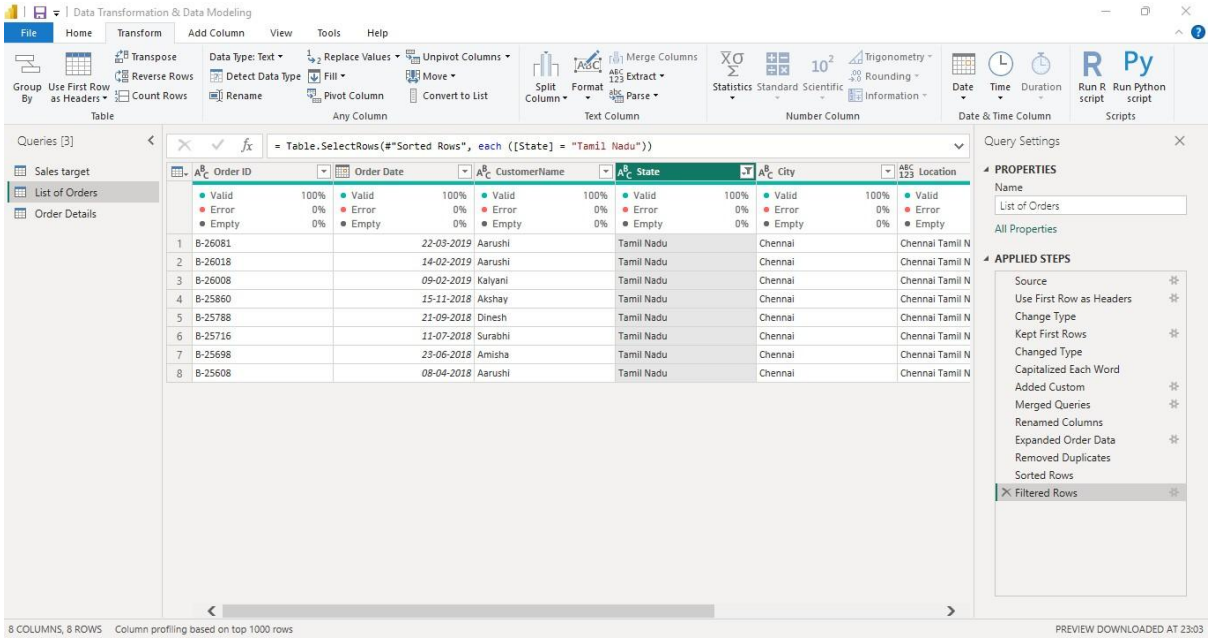
7.B) Define and apply appropriate strategies such as: Replacing missing values , Removing records , Retaining nulls with justification:



8.A) Sort records in the Orders Data table by Order Date in descending order:

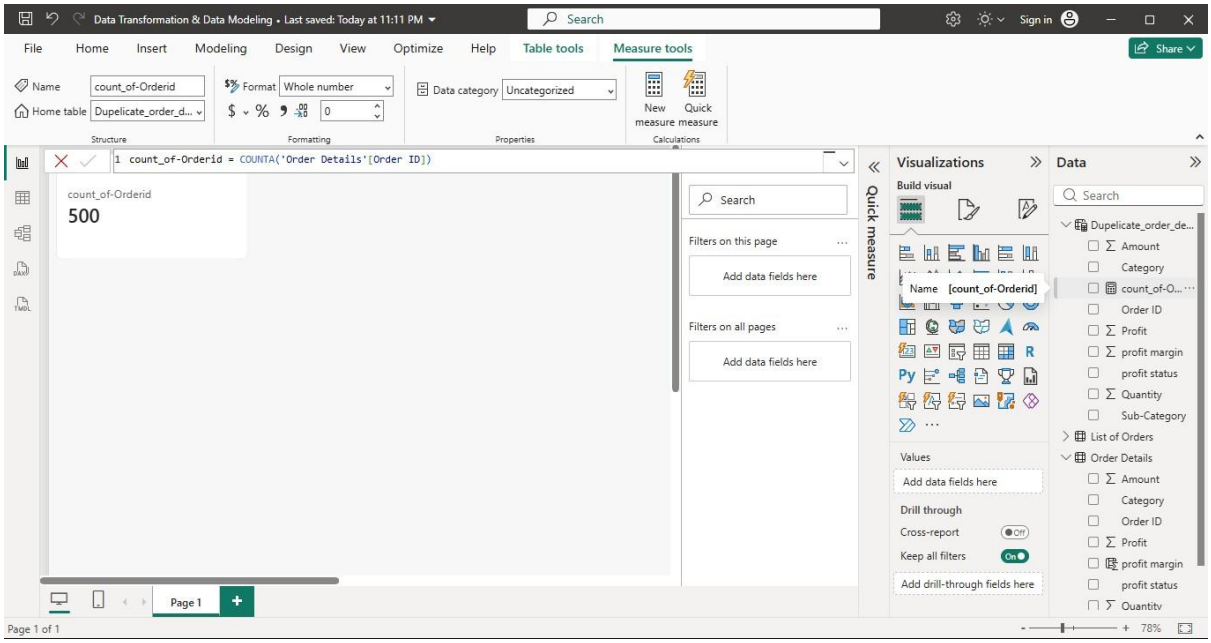


8.B) Apply filters to analyze orders from a specific state (e.g., Tamil Nadu):

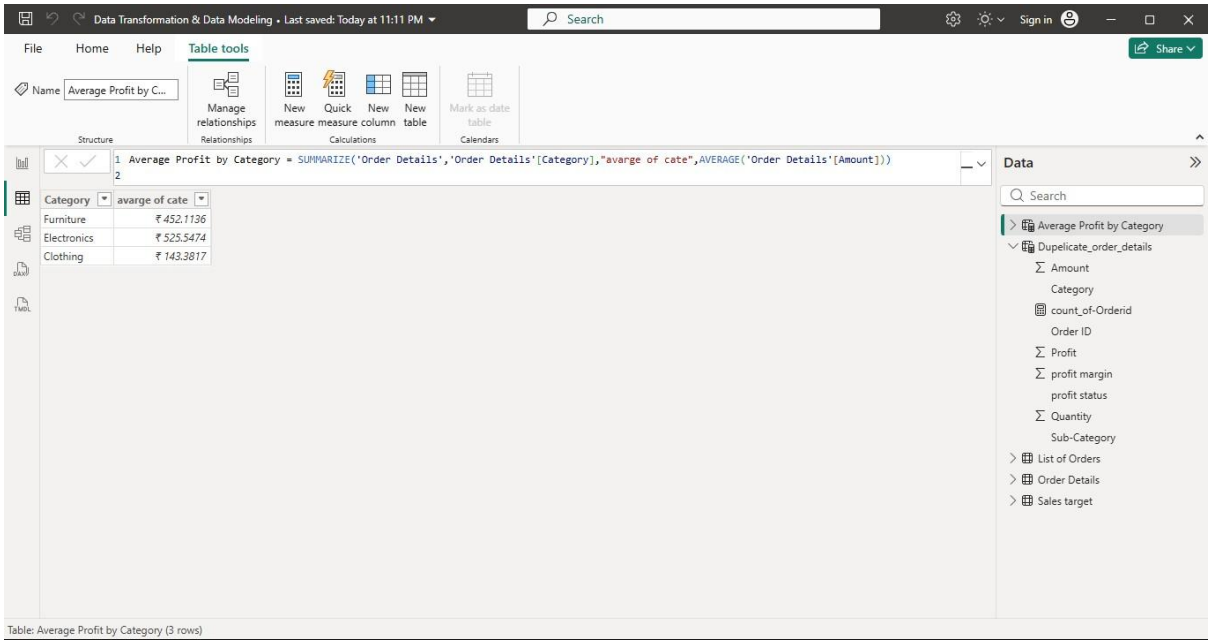


9.Duplicate the Order Details table and perform the following:

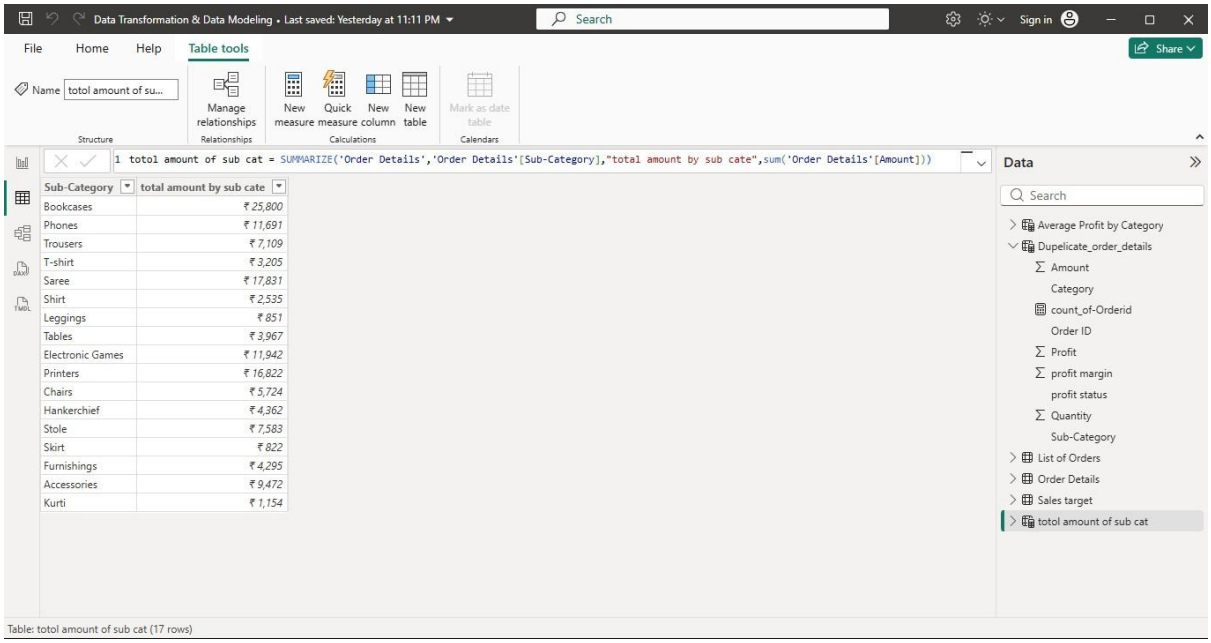
A)Count of Order ID:



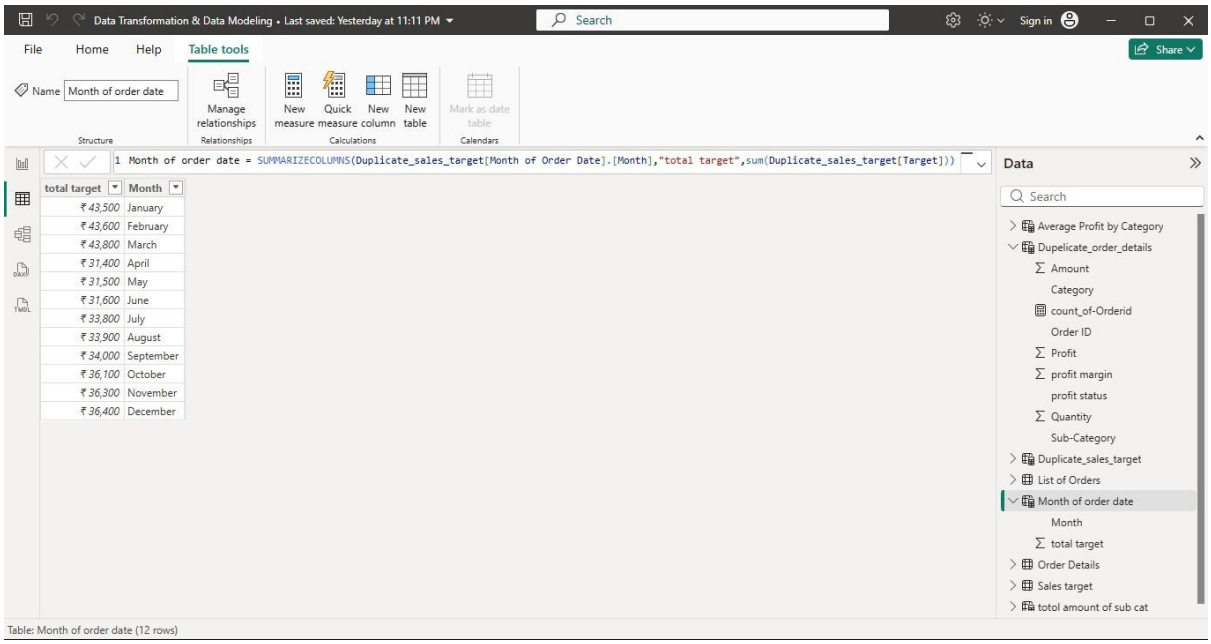
B) Average Profit by Category:



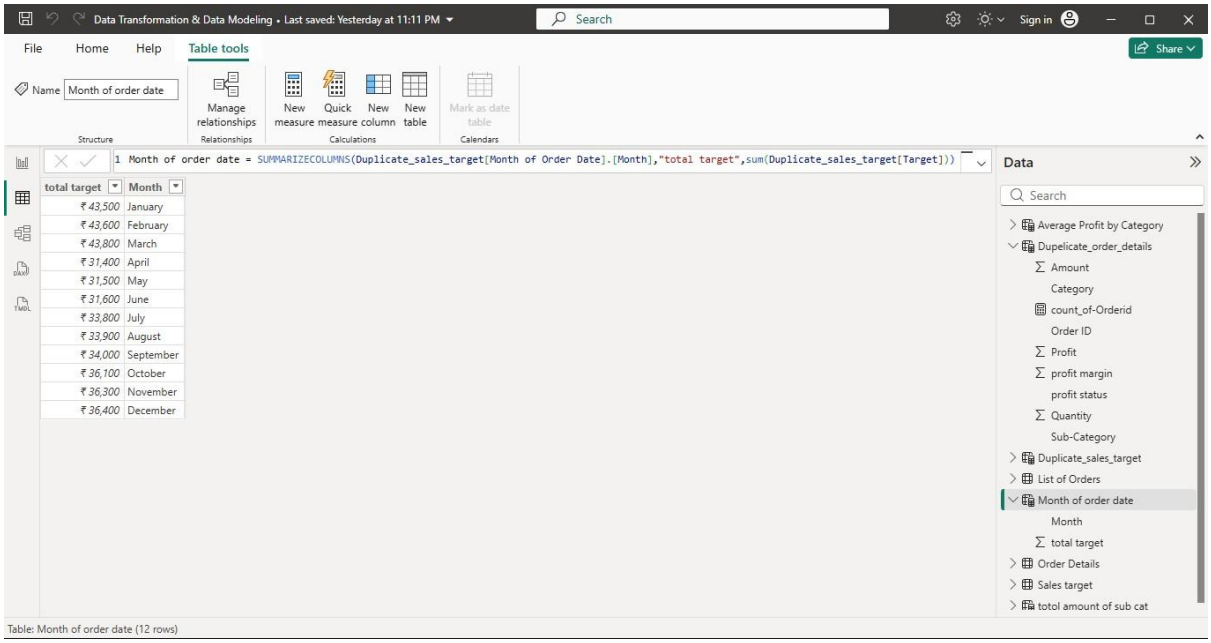
C) Total Amount by Sub-Category:



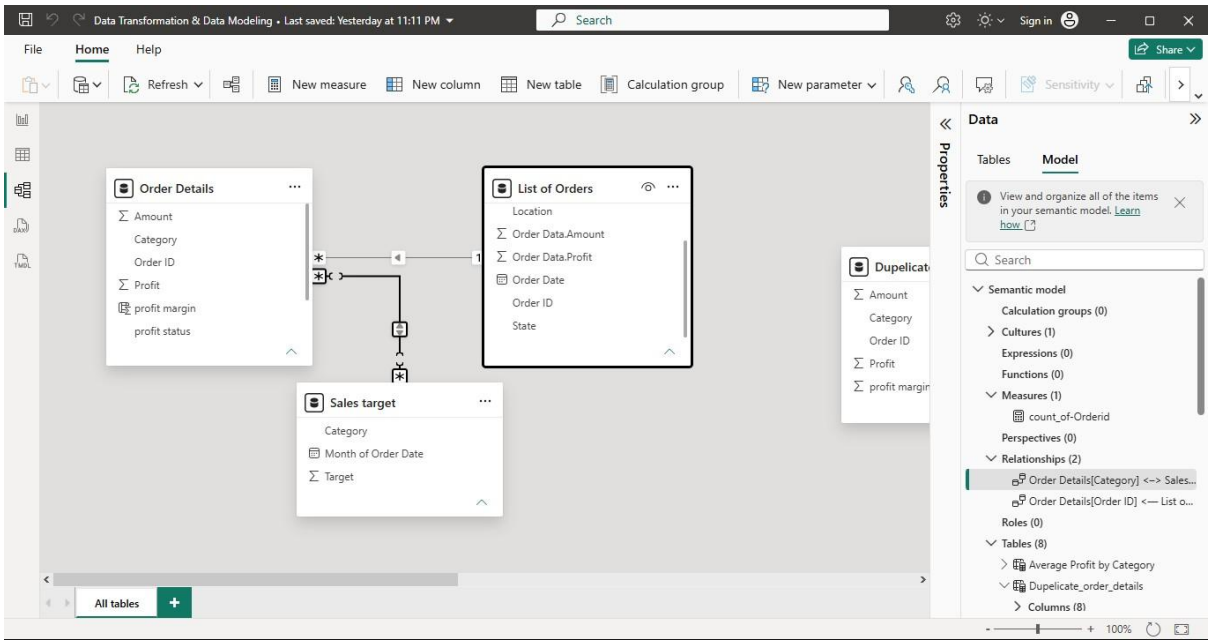
D) Duplicate the Sales Target table and aggregate the total Target by Month of order date:



D) Duplicate the Sales Target table and aggregate the total Target by Month of order date:



10. Use Manage Relationships to ensure relationships are active and correctly defined:



Conclusion:

This assignment helped me learn data cleaning, data transformation, data merging, aggregation, and data modelling in Power BI. It improved my skills in preparing data and analysing sales, profit, and sales targets.